SHOP
Shell Higher Olefins Process

Royal Dutch Shell, Netherlands

SHOP is a chemical process for the production of linear α-olefins via ethylene oligomerization and olefin metathesis.
SHOP is invented and exploited by Royal Dutch Shell

The SHOP Process

An Example of Industrial Creativity

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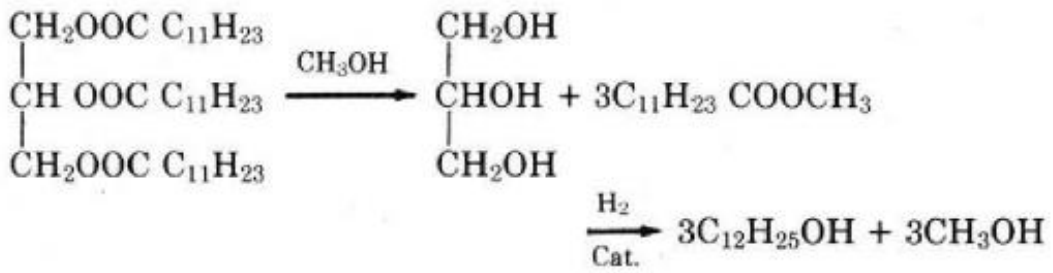
The Shell Higher Olefins Process (SHOP) is probably the most remarkable industrial chemical process to have been developed in the past decade. It involves four reactions—oligomerization, linear hydroformylation, double bond isomerization, and metathesis—reactions which have scarcely found their way into college chemistry textbooks. At the same time it has satisfied an important industrial need. The new process involves the manipulation of older reactions with a particular application in mind; it is this mission-

The chemical industry does not like agricultural feedstocks because supplies are unreliable and fluctuate with factors such as the weather and attack by pests. As the detergent industry grew, various routes to straight-chain intermediates were devised based on petrochemical feedstocks. All these suffered from the same problem; they led to mixtures of products that spanned a range of molecular weights. Once the compounds in the detergent range had been extracted, byproducts of lower and higher chain length remained, of

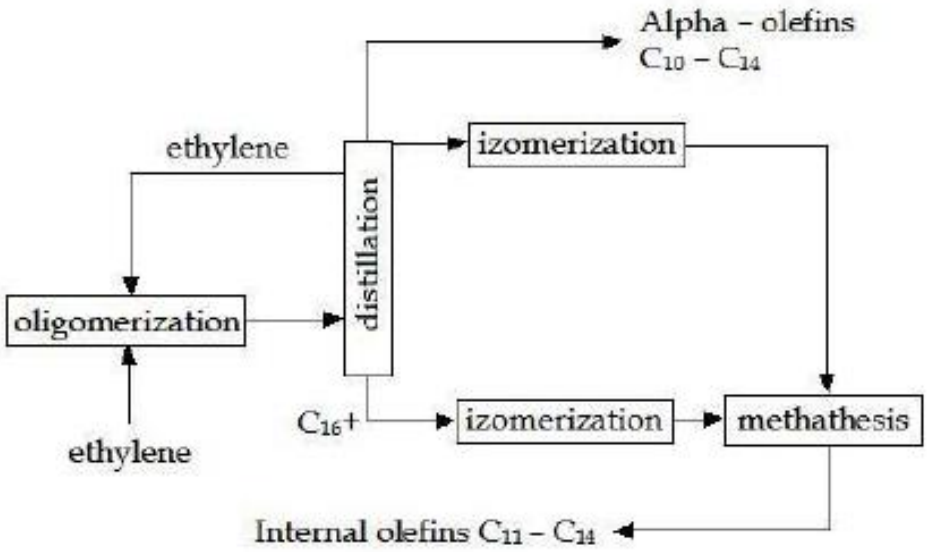


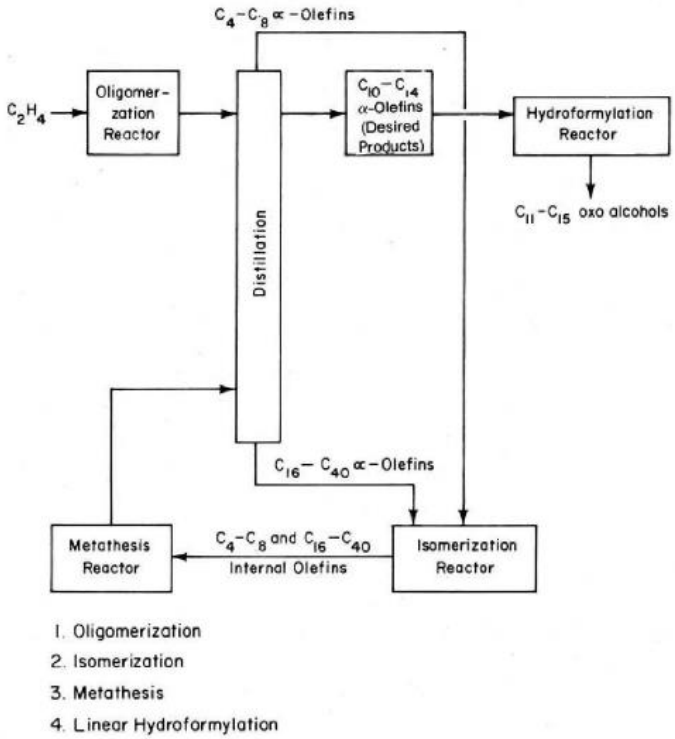
Traditional route to prepare lauryl alcohol from coconut oil

Aim of SHOP is to produce higher α–olefins and internal olefines

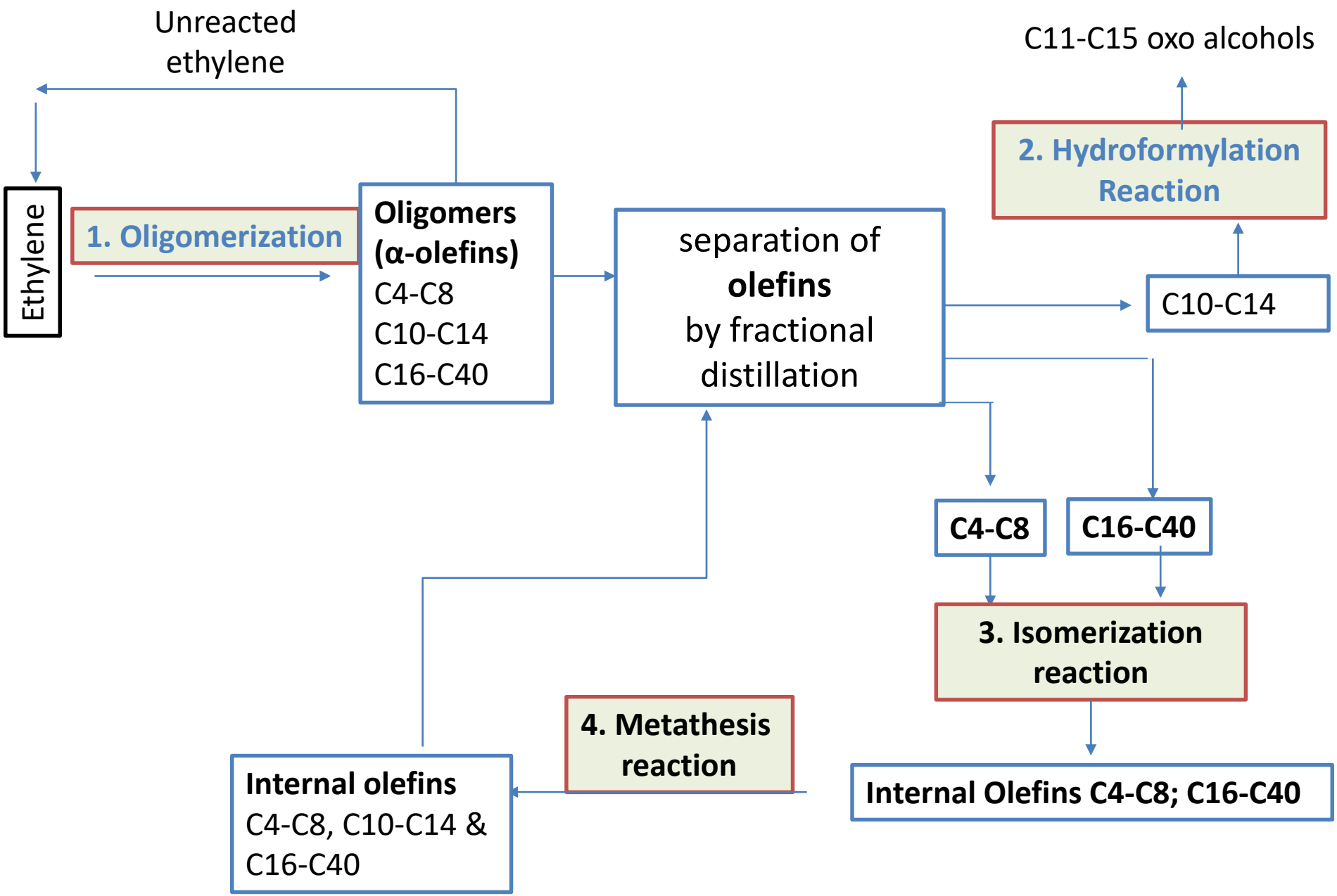
SHOP involves four major steps

1. **Oligomerization (ethylene to higher alkenes) by Ni Catalyst**
2. **Linear Hydroformylation**
3. **Double bond Isomerization on solid Potassium metal**
4. **Metathesis over alumina supported Molibdate Catalyst**



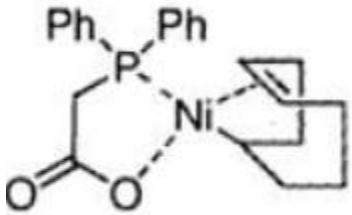


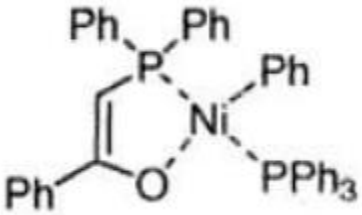
Shell higher olefin process (SHOP).



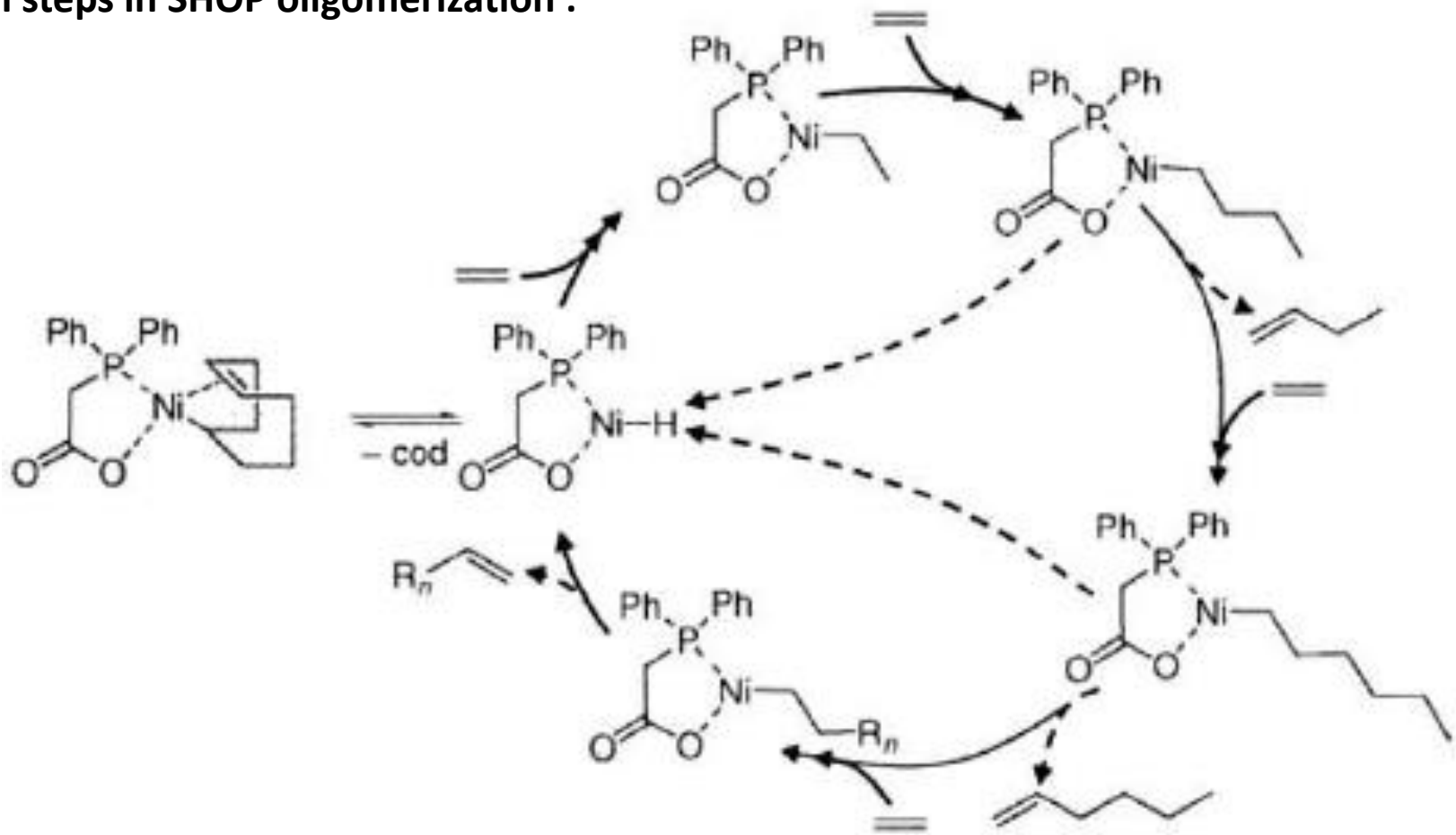
Shell Higher Olefin Process

Commonly used oligomerization catalysts:





General steps in SHOP oligomerization :



1. Oligomerization

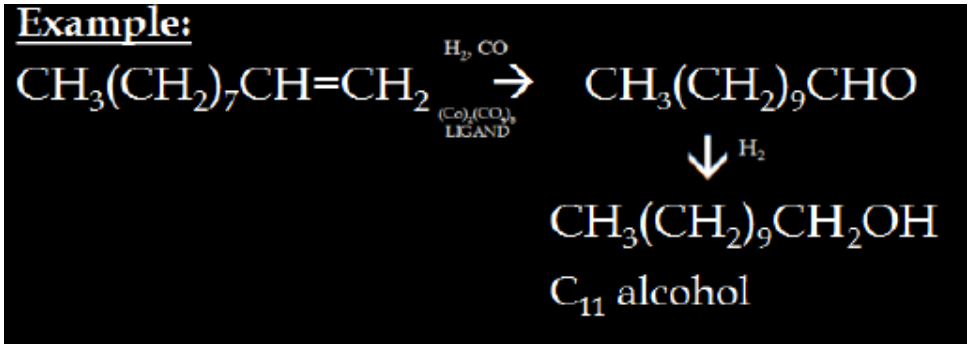
In the above SHOP oligomerization it produce linear alpha olefins ranging from C4 to C40.

They are separated by the means of **distillation.** C4 – C8 ; C10 – C14; & higher C14 α – olefins

The mixture is fractionated and the target olefins (C10-14) are collected. The light (C4-C8) and heavy (>C14) fractions (most of the mass balance) are transferred to an isomerization unit.

2. Linear Hydroformylation

The linear α –olefins (C10 – C14 α– olefins) will be converted into C11 – C15 alcohols by hydroformylation reactions. These alcohols will be used in detergent preparation



NOTE: Why linear fatty alcohols? Detergents made out of branched fatty alcohols take long time for biodegradation. So by keeping environment concern in mind linear fatty alcohols (which will bio degrade much faster) are started using in detergent preparation

3). Double bond Isomerization:

The mixture of lower and higher linear α – olefins is passed over an olefin isomerization catalyst to give a statistical mixture of linear internal olefins, which are then carried forward to a metathesis unit.

Isomerization of $C_4 - C_8$ and $C_{16} - C_{40}$ α - olefins

- heterogenous catalyst:

Na/K on Al_2O_3 or a MgO in the liquid phase

$C_4 - C_8$ α – olefins → $C_4 - C_8$ internal olefins

$C_{16} - C_{40}$ α – olefins → $C_{16} - C_{40}$ internal olefins

3) Metathesis

The mixed olefins are subjected to metathesis with a supported molybdenum catalyst. This gives a broad mixture of linear internal olefins, of which 10-15% are in the desired range. These are collected from the stream and the undesired fractions are recycled back to be isomerized.

Metathesis of $C_4 - C_8$ and $C_{16} - C_{40}$ internal olefins

- Short and long chain internal olefins disproportionate

$$CH_3CH{=}CHCH_3 + CH_3(CH_2)_8CH{=}CH(CH_2)_8CH_3 \rightarrow$$

C_4 Internal olefin $\qquad$ C_{20} internal olefin

$$\rightarrow 2\ CH_3CH{=}CH(CH_2)_8CH_3$$

C_{12} internal olefin

Fractionation of internal olefins

- $C_{10} - C_{14}$ internal olefins
- $C_4 - C_8$ internal olefins
- $C_{16} - C_{40}$ internal olefins

$C_4 - C_8$ and $C_{16} - C_{40}$
are again subjected to metathesis

Linear hydroformylation of $C_{10} - C_{14}$ internal olefins

$CH_3(CH_2)_8CH{=}CH\text{-}CH_3 \rightarrow$

C_{12} internal olefin

$CH_3(CH_2)_8CH_2CH_2CH_2CHO \rightarrow$

$CH_3(CH_2)_{11}CH_2CH_2OH$

C_{14} alcohol

Flexibility of SHOP process

- At each step we can decide to stop further „processing" of olefins, adjusting production to present needs.

Here cobalt catalyst will be used for hydroformylation of internal olefins. These cobalt catalyst will isomerize Internal olefins to alpha olefins and then hydroformylation proceeds to desired alcohols. Such an isomerization wont happen with Rhodium catalyst.

C4 – C10 α – olefins: used as **monomers** for the production of polymers

C6 – C10 α– olefins: will be used as **plasticizers** for plastics, cement, concrete, wallboard and clay bodies.

C12 – C20 α– olefins : used for the preparation of linear fatty alcohols for preparing Biodegradable detergents